

Polynomial conditioning and rational ranks for two-variable ordered prefixes

Proof independently reviewed by size_bounds

30 September 2026

Write $w = t(1 - t)$, $H_d(t) = \int_0^t P_d(2u - 1)du$, and $R_i(t) = t - \eta_i H_{d_i}(t)$, where $0 < \eta_i \leq 1/2$. The increasing functions R_i are distribution functions. Define

$$J(t) = \int_0^t R_1(u)R_2'(u) du.$$

Thus $J(t)$ is the probability that independent variables with these CDFs satisfy $X_1 < X_2 < t$. Its reversed counterpart is $R_1 R_2 - J$.

Lemma 1. *Let $r \geq 0$ and choose*

$$d_1 > 2(r + 2) + 1, \quad d_2 > 2(d_1 + r + 3) + 1, \quad D = r + d_1 + d_2 + 2.$$

For $p_0, p_1, p_2, p_{12}, p_J \in \mathcal{P}_r$,

$$(\|p_0\|_2^2 + \|p_1\|_2^2 + \|p_2\|_2^2 + \|p_{12}\|_2^2 + \|p_J\|_2^2)^{1/2} \leq C(D+1)^{20}(\eta_1\eta_2)^{-2}\|p_0+p_1R_1+p_2R_2+p_{12}R_1R_2+p_JJ\|_2.$$

All norms use Lebesgue measure on $[0, 1]$.

Proof. We use the exact primitive and trinary-band lemmas from `ordered_prefix_legendre_tools.tex`. Set $\lambda_1 = d_1(d_1 + 1)$ and $I_1(t) = \int_0^t H_{d_1}(u)du$. The primitive formula gives

$$I_1 = \frac{w'H_{d_1} - wH_{d_1}'}{\lambda_1 - 2}.$$

Expanding J yields

$$J = t^2/2 - \eta_1 I_1 - \eta_2 \int_0^t uH_{d_2}'(u)du + \eta_1\eta_2 \int_0^t H_{d_1}(u)H_{d_2}'(u)du.$$

The last two terms have the form $AH_{d_2} + BH_{d_2}'$ with $\deg A, \deg B \leq d_1 + 2$. This follows from the primitive lemma with coefficient polynomials u and H_{d_1} , respectively. The other terms containing H_{d_2} in the stated linear combination have coefficient degrees no larger. After multiplication by the p 's, all such high-band terms are orthogonal to $\mathcal{P}_{d_1+r+2}$ by the choice of d_2 .

Consequently the orthogonal projection L of the full combination F onto that low-degree space is exactly

$$L = p_0 + t(p_1 + p_2) + t^2 p_{12} + \frac{1}{2}t^2 p_J - \eta_1(p_1 + tp_{12})H_{d_1} - \eta_1 p_J I_1.$$

In its trinary representation $P + H_{d_1}Q + H_{d_1}'V$, the coefficient

$$V = \frac{\eta_1}{\lambda_1 - 2}wp_J$$

is unique, and all three coefficient degrees are at most $r + 2$. Trinary coercivity bounds $\|V\|_2 \leq C(D + 1)^6 \|L\|_2$. For a polynomial $p \in \mathcal{P}_r$,

$$\|wp\|_2 \geq c(r + 1)^{-2} \|p\|_2. \quad (1)$$

Indeed, exclude endpoint intervals of length $c(r + 1)^{-2}$; the polynomial evaluation bound preserves at least half the squared norm, and w is at least $c(r + 1)^{-2}$ on what remains. It follows that

$$\|p_J\|_2 \leq C(D + 1)^{10} \eta_1^{-1} \|F\|_2.$$

Subtract $p_J J$. Since $0 \leq J \leq 1$, the remaining combination has norm at most $(1 + C(D + 1)^{10} \eta_1^{-1}) \|F\|_2$. The two-band rank-product conditioning theorem from `lacunary_legendre_conditioning.tex` bounds its four coefficients by $C(D + 1)^{10} (\eta_1 \eta_2)^{-1}$ times this norm. The displayed bound follows, after increasing the absolute constant. $\square$

Theorem 1 (Rational ordered-prefix data). *For every fixed $T \geq 1$ and every $m \geq 1$, there are $K = O((m + 1)^4)$ separated ordered real nodes $b_q \in (0, 1)$ and rational arrays $r_{1,q}, r_{2,q}, a_q$ with a common denominator bounded by a fixed power of $m + 1$ (depending on T), such that for $0 \leq j \leq m$,*

$$\begin{aligned} K^{-1} \sum_q b_q^j &= \frac{1}{j + 1}, & K^{-1} \sum_q b_q^j r_{i,q} &= \frac{1}{j + 2} \quad (i = 1, 2), \\ K^{-1} \sum_q b_q^j r_{1,q} r_{2,q} &= \frac{1}{j + 3}, & K^{-1} \sum_q b_q^j a_q &= \frac{1}{2(j + 3)}. \end{aligned}$$

The node and rank gaps, including their endpoint gaps, are at least c/K , and $|r_{i,q} - b_q| \leq C_T(m + 1)^{-T}$. Furthermore these rational arrays are locally consistent ordered-prefix data: append

$$(r_{1,0}, r_{2,0}, a_0) = (0, 0, 0), \quad (r_{1,K+1}, r_{2,K+1}, a_{K+1}) = (1, 1, 1/2).$$

For every $0 \leq q \leq K$,

$$0 < a_{q+1} - a_q - r_{1,q}(r_{2,q+1} - r_{2,q}) < (r_{1,q+1} - r_{1,q})(r_{2,q+1} - r_{2,q}). \quad (2)$$

This does not assert a full fair family with three small dice.

Proof. Choose $d_1, d_2 = O(m + 1)$ satisfying Lemma 1 for $r = m - 1$, and, if needed, increase their absolute additive constants so that $d_1 \geq m + 3$ and $d_2 > m + d_1 + 3$. Choose $\eta_1 = \eta_2 = c(m + 1)^{-(T+1)}$. Legendre orthogonality gives, for $0 \leq j \leq m$,

$$\int t^j R_i(t) dt = \frac{1}{j + 2}, \quad \int t^j R_1(t) R_2(t) dt = \frac{1}{j + 3}, \quad \int t^j J(t) dt = \frac{1}{2(j + 3)}.$$

For the last identity, integration by parts gives $[J(1) - \int t^{j+1} R_1 R_2'] / (j + 1)$; orthogonality gives $J(1) = 1/2$ and the other integral $1/(j + 3)$. Take a real equal-weight quadrature of sufficiently high degree $O(m)$ with $K = O(m^4)$ near-midpoint nodes b_q^0 , as in the reviewed separated quadrature lemma. Its degree is at least twice the maximal degree of a polynomial in Lemma 1, ensuring exact Gram matrices as well as all the displayed moment identities.

Round and correct the two rank columns to polynomial-denominator rationals, preserving exactly their three constant mixed moments, by the disjoint-row construction in `../independent_directions/multirank_step_compatibility.tex`. The correction from the real baseline can be made $O((m + 1)^{-A})$ for any fixed A by increasing the initial polynomial denominator. Independently round the $a_q = J(b_q^0)$ to a polynomial grid whose denominator is a multiple of six and of the rank denominator. Flooring and adding one grid unit to a suitable number of entries preserves their exact mean $1/6$, with at most one grid unit error per entry.

Keep all rational arrays fixed, and correct the nodes to restore the nonconstant moment equations. In an antiderivative Legendre basis, the baseline Jacobian evaluates ϕ_j times $1, R_1, R_2, R_1 R_2, J$, $0 \leq j < m$. Its empirical Gram is exact, and Lemma 1 gives a polynomial least singular value. All derivative bounds are polynomial. Thus the same fixed-inverse Newton contraction used in the multirank theorem moves the nodes by $O((m+1)^{-A'})$, with arbitrarily prescribed fixed A' , if the initial denominator exponent is sufficiently large. This keeps all gaps and gives the stated exact moments and closeness.

Finally, at the continuous baseline the middle expression in (2) equals

$$\int_{b_q^0}^{b_{q+1}^0} [R_1(t) - R_1(b_q^0)] R_2'(t) dt.$$

Both this and its complement to the product of the two rank increments are at least $c(b_{q+1}^0 - b_q^0)^2 \geq c/K^2$, since $1/2 \leq R_i' \leq 3/2$. Include the endpoint intervals in this argument. Errors much smaller than K^{-2} preserve the strict inequalities. Moving the nodes does not change them. All rank arrays retain one polynomial common denominator. $\square$

A related first-moment variant. The same argument conditions $1, R, M$, where $R = t - \eta H_d$ and

$$M(t) = \int_0^t u R'(u) du = t^2/2 - \eta \left[t H_d - \frac{w' H_d - w H_d'}{d(d+1) - 2} \right].$$

The unique H_d' coefficient again isolates the coefficient of M via (1). This can rationalize both cumulative mass and cumulative first moment while preserving their mixed scalar moments. It does not force the corrected real nodes to have polynomial rational denominators, nor does it give positive completion with additional independent variables.